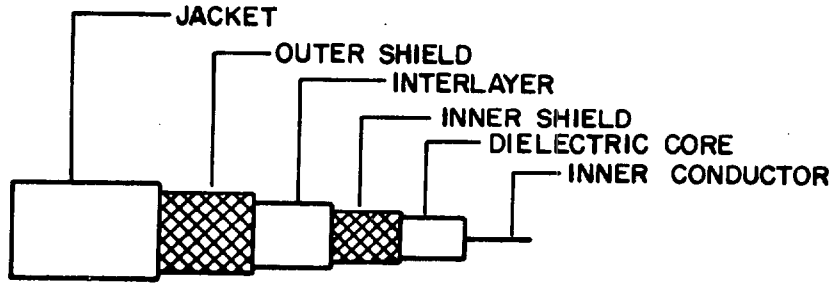
MILITARY SPECIFICATION SHEET

# CABLE, RADIO FREQUENCY, TRIAXIAL, 0.245 INCH, 50 OHM, WATERBLOCKED, NON-WATER BLOCKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.




CABLE SHOULD BE CONSTRUCTED OF NON-TOXIC MATERIAL

FIGURE 1. Triaxial cable, radio frequency M17/134-00001.

Ⓑ denotes changes

FSC 6145

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TABLE I. Description.

| Component | Construction |
|---|---|
| Inner conductor | Solid strand of silver-coated copper wire.<br>Diameter: 0.033 inch ±0.001 |
| Dielectric core | Type A-1: Solid polyethylene<br>Diameter: 0.116 inch ±0.004 |
| Inner shield | Single braid of AWG No. 36 silver-coated copper wire.<br>Diameter: 0.150 inch maximum<br>Coverage: 90% minimum<br>Carriers: 16<br>Ends: 6<br>Picks/inch: 10.4<br>Separater tape optional |
| Interlayer | Type A-1: Solid polyethylene<br>Diameter: 0.175 inch ±0.005 |
| Outer shield | Single braid of AWG No. 36 silver-coated copper wire.<br>Diameter: 0.203 inch maximum<br>Coverage: 90% minimum<br>Carriers: 16<br>Ends: 8<br>Picks/inch: 9.8<br>Separater tape optional |
| Jacket | Type XII: Polyurethane<br>Diameter: 0.245 inch ±0.005 |

Note: (Free strippable blocking compound needed where necessary to meet hydrostatic test. The inner shield may be partially imbedded in the dielectric. Applicable to water blocked cable only).

ENGINEERING INFORMATION:

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +70°C.

Inner conductor properties:

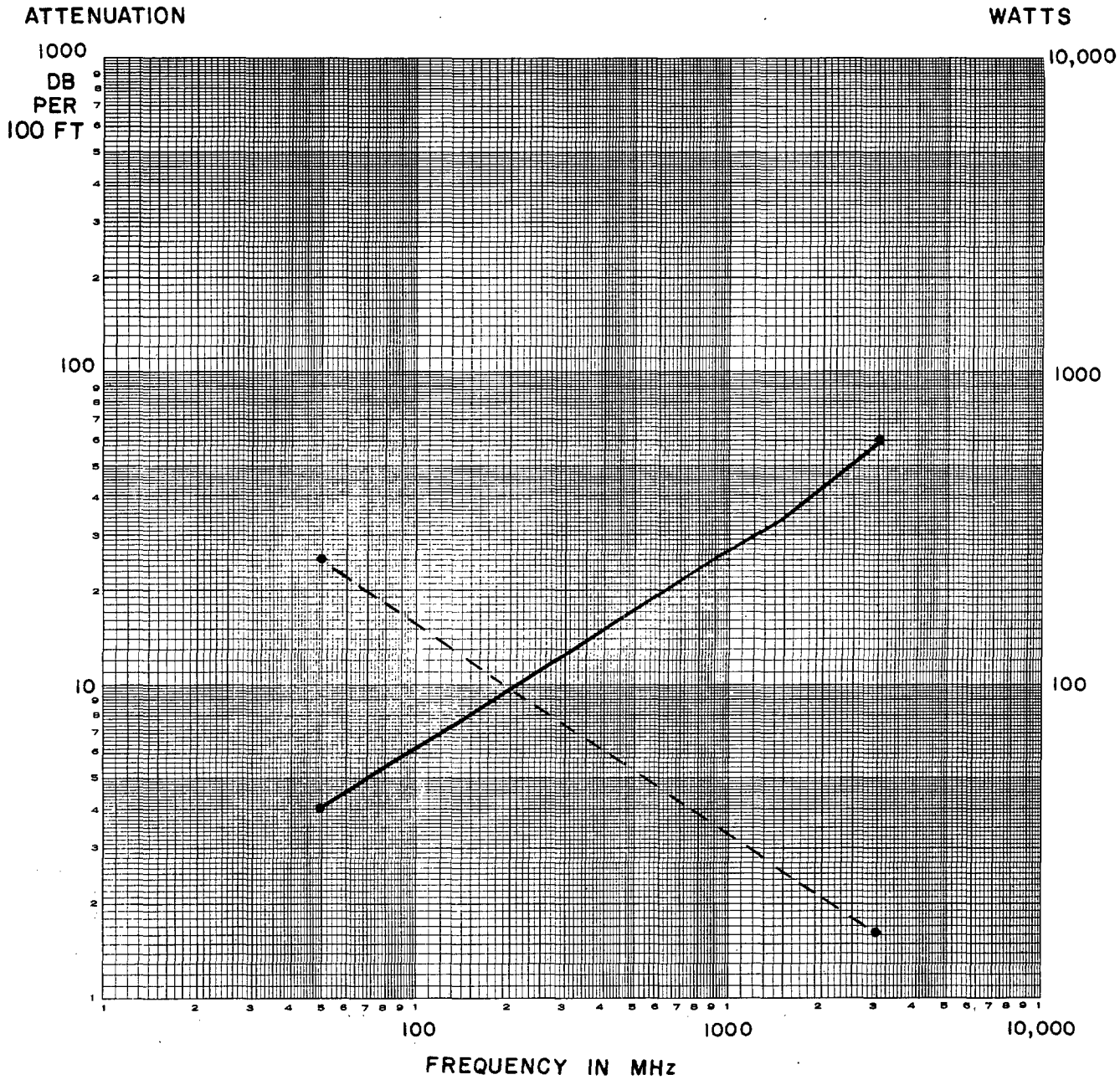
DC resistance (maximum at 20°C): 1.05 ohms per 100 feet.

(B) Elongation: 25 percent, minimum.

Tensile strength: Not applicable.



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Maximum power — — — — at 25°C sea level

Maximum attenuation ————————

| Frequency MHz | Attenuation dB | Watts dB |
|---|---|---|
| 50 | 4 | 240 |
| 100 | 6 | 150 |
| 1000 | 26 | 33 |
| 3000 | 60 | 16 |

(Test requirements shall be as line indicated on graph)

FIGURE 2. Power rating and attenuation.



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Engineering note: This cable useful for nonhosing low temperature applications .

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

(B) Eccentricity: 10 percent, maximum.

Adhesion of conductors:

M17/134-00001: 2 pounds minimum, 20 pounds maximum.
M17/134-00002: 2 pounds minimum, 20 pounds maximum.

Aging stability: 90°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(B) Cold bend: -55°C ±2°C.

Dimensional stability:

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 4.5 pounds per 100 feet, maximum.

Watertightness (Not applicable to M17/134-0002): One end of a five foot length of completed cable shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 $lb_f/in^2$ for a period of 6 hours. Any evidence of water leakage shall be a cause for rejection.

Hydrostatic (open end) (Not applicable to M17/134-0002): A 5-foot specimen of completed cable shall be fitted into a gland similar to USL 44500B-12. The cable shall be capable of withstanding 1,000 $lb_f/in^2$ applied to the gland end of the specimen for a period of 2 hours. Any evidence of water leakage shall be cause for rejection.

Electrical:

Continuity: Applicable.

(B) Spark test: 5,000 V rms, +10%, -0%.

Voltage withstanding: 5,000 V rms minimum applied between the inner and outer conductors with the outer conductor grounded; 500 V dc minimum applied between the outer conductor and the extra shield with the extra shield grounded.



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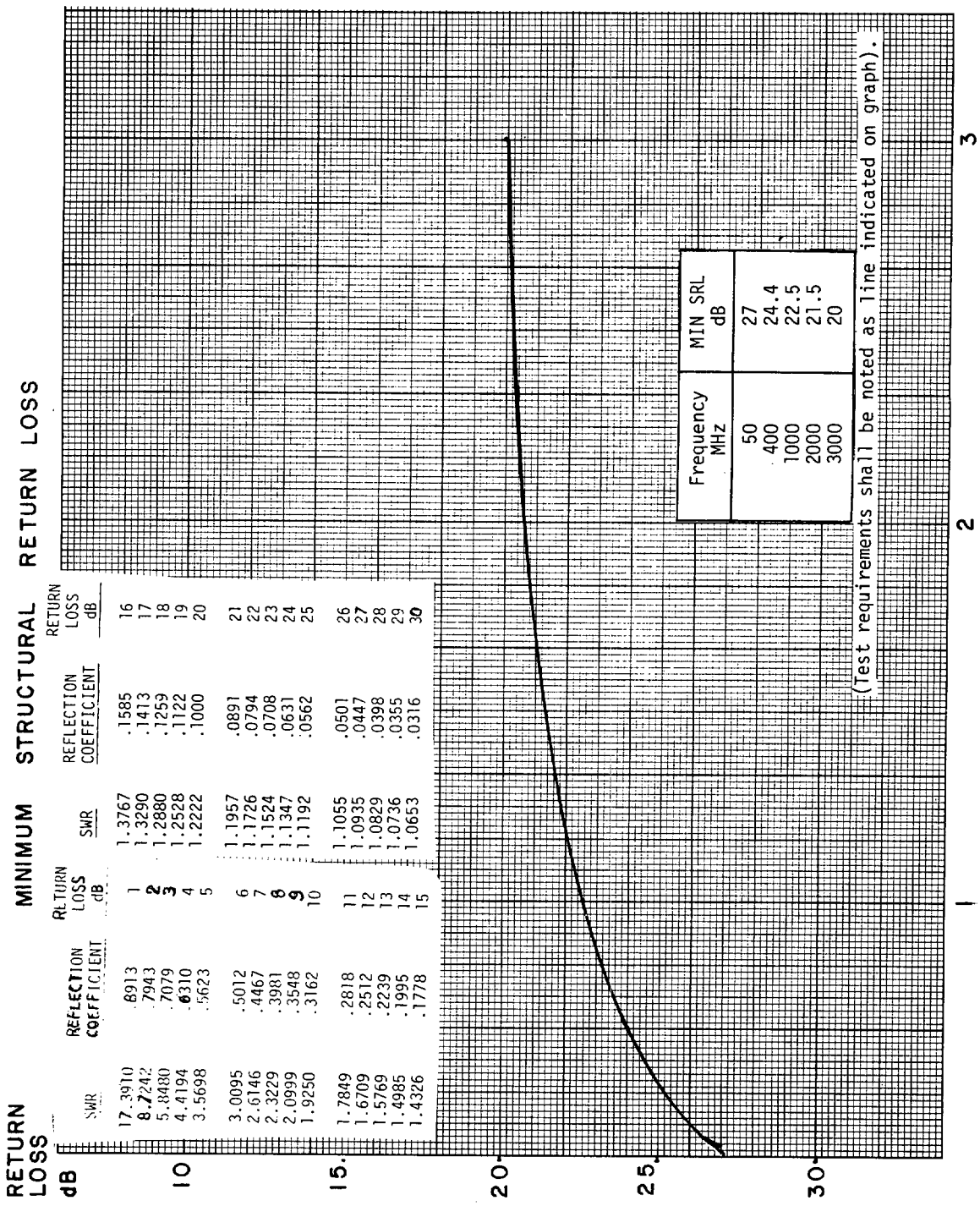



FIGURE 3. Structural return loss.



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Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

(B) Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

| Part number | Superseded part number or type designation | Type |
|---|---|---|
| M17/134-00001 | TRF58 | Water blocked |
| M17/134-00002 | YR19210 | Non-water blocked |

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 6145-0911-28)

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